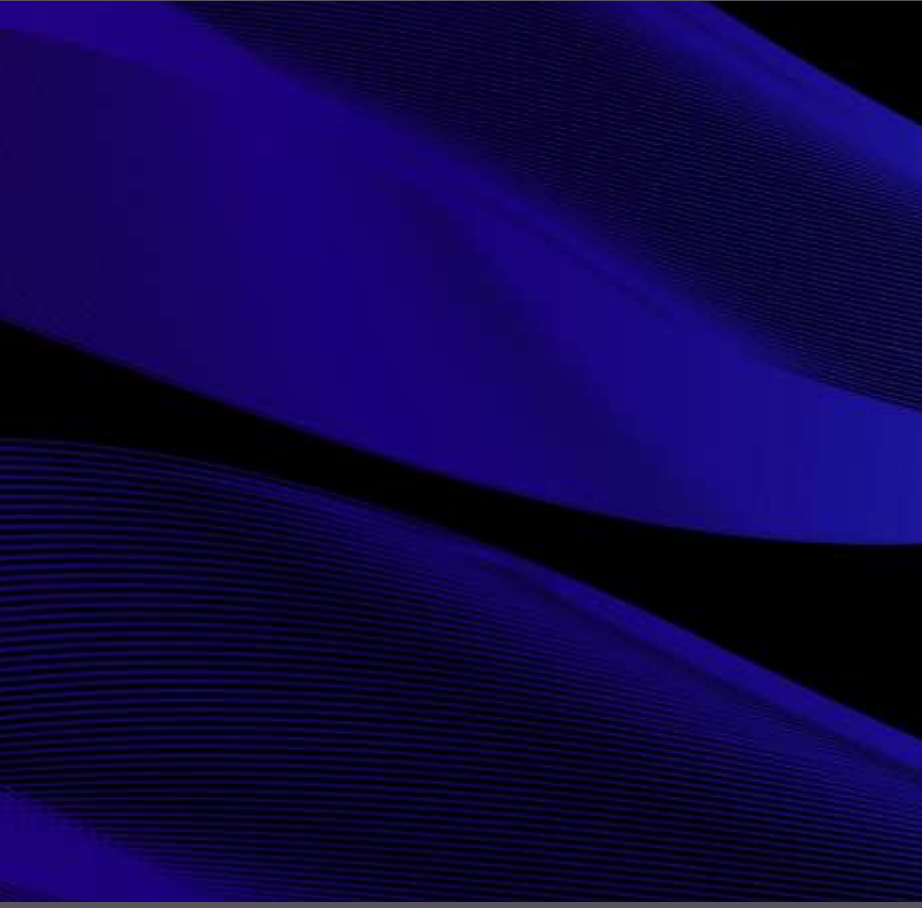




SYSTEM CONTROLLING USING HAND GESTURE

Mediapipe

Hand

Landmark

Indexing

Mediapipe labels each finger joint with a specific index number. Below is the landmark mapping:

Landmark Index	Finger	Position
0	Wrist	Base of the hand
1	Thumb	First joint
2	Thumb	Second joint
3	Thumb	Third joint
4	Thumb	Tip (important for gestures)
5	Index Finger	First joint
6	Index Finger	Second joint
7	Index Finger	Third joint
8	Index Finger	Tip (important for gestures)
9	Middle Finger	First joint
10	Middle Finger	Second joint
11	Middle Finger	Third joint
12	Middle Finger	Tip (important for gestures)
13-20	Ring & Pinky Fingers	Similar structure

Calculating Finger Distances

```
distance_thumb_index = get_finger_distance(lmList, 4, 8)  
distance_index_middle = get_finger_distance(lmList, 8, 12)
```

Calls a function `get_finger_distance()` to compute the Euclidean distance between:

Thumb (4) and Index (8) → `distance_thumb_index`

Index (8) and Middle (12) → `distance_index_middle`

Gesture-Based Controls

Mouse Control

```
if distance_index_middle > 0.1:
```

```
    pyautogui.moveTo(x_index, y_index,  
duration=0.1)
```

```
if distance_thumb_index < 0.05:
```

```
    pyautogui.click()
```

Moves the cursor to the index finger position if the index-middle distance is large.

Clicks the mouse if the thumb-index distance is very small (indicating a pinch)

Volume Control

```
if distance_index_middle < 0.05 and  
distance_thumb_index > 0.1:  
    volume_value =  
    np.interp(distance_thumb_index, [0.05, 0.2], [-  
65, 0])  
    volume.SetMasterVolumeLevel(volume_value,  
None)
```

If index and middle fingers are close, but thumb and index are apart, adjust volume.

Uses `np.interp()` to map the thumb-index distance to volume levels between -65 dB (mute) and 0 dB (max).

Brightness Control

```
if distance_thumb_index > 0.05 and  
distance_thumb_index < 0.2:  
    brightness_value =  
    np.interp(distance_thumb_index, [0.05, 0.2], [10,  
100])  
  
sbcontrol.set_brightness(int(brightness_value))
```

If thumb-index distance is moderate, adjust screen brightness.

Uses `np.interp()` to map thumb-index distance to brightness levels between 10% and 100%.

THANKYOU !

